

Supplementary information of

Open fire exposure increased the risk of pregnancy loss in South Asia

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Supplemental Tables

Supplementary Table 1 Population characteristics of the analyzed samples.

Variable	Group	Pregnancy loss	Miscarriage	Stillbirth
		Samples (N)		
Matched number of controls	1	10328	8075	2253
	2	8005	6537	1468
	3	3724	2940	784
	4	1720	1329	391
	≥5	1099	849	250
Country	Bangladesh	2864	2083	781
	India	18862	15386	3476
	Pakistan	3150	2261	889
Education	No education	7080	5068	2012
	Primary	3947	2971	976
	Secondary or higher	13849	11691	2158
Residence*	Urban	8162	6807	1355
	Rural	16714	12923	3791
Insurance	Not covered	18787	15115	3672
	Covered	2609	2124	485
	Unknown	3480	2491	989
Employment status	Rare or never	1434	1121	313
	Occasional	532	388	144
	Seasonal	201	148	53
	All year	6722	5127	1595
	Unknown	15987	12946	3041
Anemia	No	9198	7638	1560
	Yes	9788	7822	1966
	Unknown	5890	4270	1620
Body mass index	Underweight	4259	3227	1032
	Normal	13153	10580	2573
	Overweight or obese	4975	4128	847
	Unknown	2489	1795	694
Smoking	No	2594	2102	492
	Yes	18802	15137	3665
	Unknown	3480	2491	989
		Mean (standard deviation)		
Fire PM _{2.5} (µg/m ³)	Case	1.30 (2.41)	1.35 (2.60)	1.11 (1.49)
	Control	1.14 (1.34)	1.17 (1.35)	1.03 (1.28)
Total PM _{2.5} (µg/m ³)	Case	53.24 (26.57)	53.28 (27.50)	53.09 (22.67)
	Control	53.19 (25.32)	53.30 (25.92)	52.76 (22.89)

Fire emission (g dry-matter/m ³ /month)	Case	0.98 (8.37)	1.10 (9.32)	0.54 (2.33)
	Control	0.80 (4.10)	0.85 (4.34)	0.59 (2.97)
Satellite burned area (%)	Case	0.14 (1.83)	0.15 (1.97)	0.09 (1.18)
	Control	0.09 (1.48)	0.10 (1.52)	0.07 (1.29)
Temperature (K)	Case	297.12 (6.31)	297.07 (6.61)	297.29 (5.01)
	Control	296.98 (6.50)	296.92 (6.73)	297.19 (5.55)
Humidity (g/kg)	Case	11.12 (4.79)	11.07 (5.05)	11.34 (3.60)
	Control	10.99 (4.93)	10.99 (5.19)	10.96 (3.75)
Age (years)	Case	26.15 (5.76)	26.31 (5.74)	25.55 (5.79)
	Control	24.48 (5.06)	24.49 (5.03)	24.45 (5.17)

* In DHS surveys, the definition of urban-rural is questionnaire-based and country-specific.

Supplementary Table 2 Response rates for the surveys incorporated into this study.

Country	DHS phase*	Response rate of the eligible woman (%)		
		Urban	Rural	Total
India	7 ¹	95.8	97.0	96.7
Pakistan	5 ²	93.3	95.3	94.5
	7 ³	93.2	95.3	94.3
Bangladesh	4 ⁴	98.3	98.8	98.6
	5 ⁵	98.1	98.5	98.4
	6 ⁶	97.0	98.4	97.9
	7 ⁷	97.5	98.1	97.9

* The rates are collected from the following sources:

1. International Institute for Population Sciences - IIPS/India, ICF. India National Family Health Survey NFHS-4 2015-16.). IIPS and ICF (2017).
2. National Institute of Population Studies - NIPS/Pakistan, Macro International. Pakistan Demographic and Health Survey 2006-07.). NIPS/Pakistan and Macro International (2008).
3. National Institute of Population Studies - NIPS/Pakistan, ICF. Pakistan Demographic and Health Survey 2017-18.). NIPS/Pakistan and ICF (2019).
4. National Institute of Population Research Training - NIORT/Bangladesh, Mitra, Associates/Bangladesh, ORC Macro. Bangladesh Demographic and Health Survey 2004.). NIORT, Mitra and Associates, and ORC Macro (2005).
5. National Institute of Population Research Training - NIORT/Bangladesh, Mitra, Associates/Bangladesh, Macro International. Bangladesh Demographic and Health Survey 2007.). NIORT, Mitra and Associates, and Macro International (2009).
6. National Institute of Population Research Training - NIORT/Bangladesh, Mitra, Associates/Bangladesh, ICF International. Bangladesh Demographic and Health Survey 2011.). NIORT, Mitra and Associates, and ICF International (2013).
7. National Institute of Population Research Training - NIORT/Bangladesh, Mitra, Associates, ICF International. Bangladesh Demographic and Health Survey 2014.). NIORT, Mitra and Associates, and ICF International (2016).

Supplementary Table 3 Comparison between the GEOS-Chem simulations and the satellite-based PM_{2.5} concentrations.

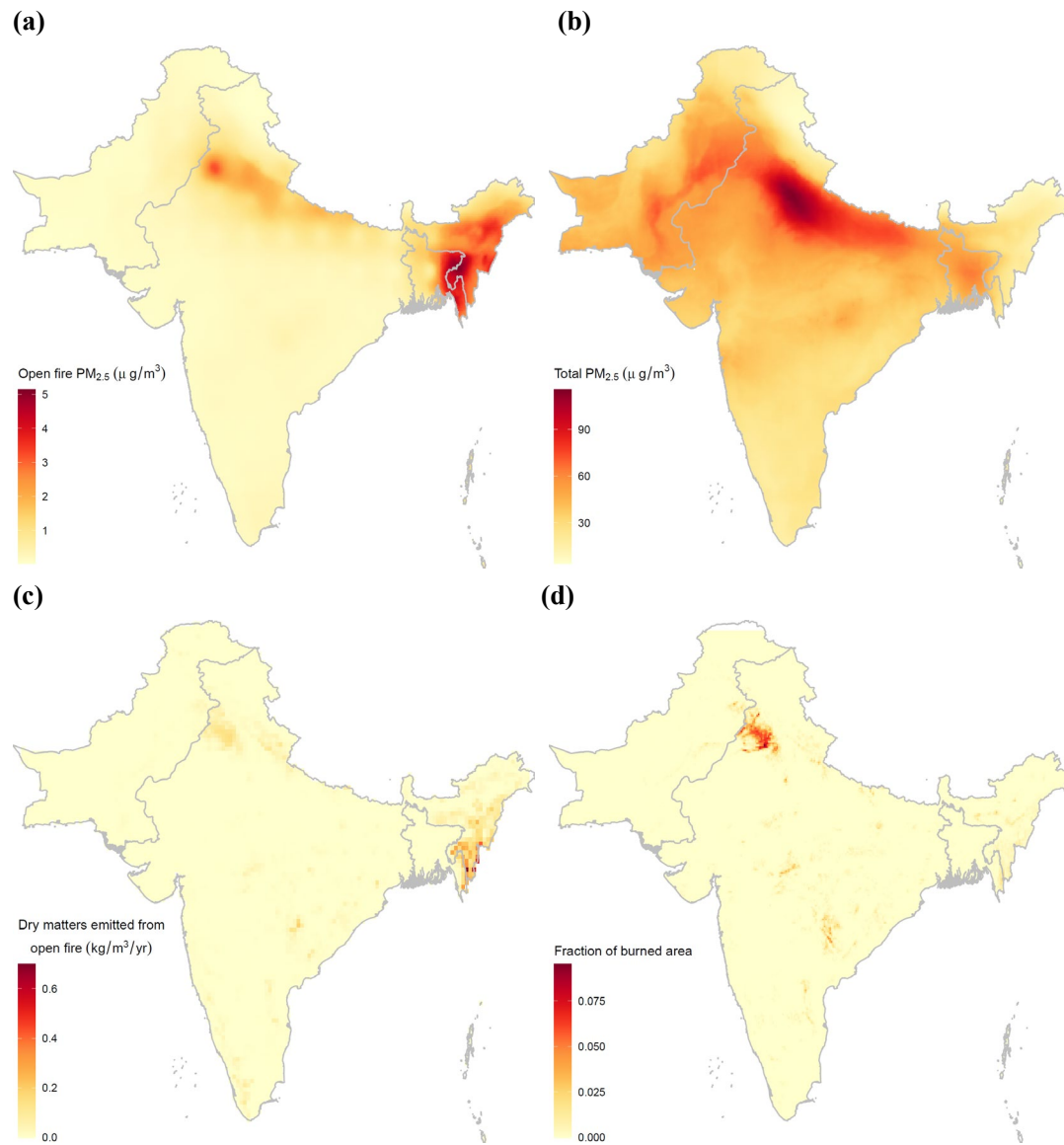
Year	Mean bias [#] (µg/m ³)	Root-squared-mean error (µg/m ³)	Regression*		Correlation R ² †
			Intercept	Slope	
2000	7.64	14.09	5.99	1.10	0.53
2001	10.43	16.90	9.84	1.03	0.47
2002	10.05	16.31	9.53	1.03	0.50
2003	10.78	17.38	9.92	1.05	0.50
2004	9.68	16.52	10.64	0.95	0.51
2005	9.04	15.00	11.29	0.88	0.50
2006	8.67	15.74	11.94	0.84	0.48
2007	8.03	15.18	11.59	0.83	0.51
2008	10.48	17.20	14.61	0.82	0.55
2009	7.46	14.07	12.26	0.79	0.60
2010	8.43	16.21	12.40	0.82	0.52
2011	6.25	16.07	13.69	0.68	0.49
2012	6.43	14.54	13.30	0.71	0.54
2013	8.23	15.99	14.79	0.73	0.53
2014	6.84	14.37	13.57	0.74	0.60

* The regression is performed as the GEOS-chem simulations against the satellite-based PM_{2.5} concentrations.

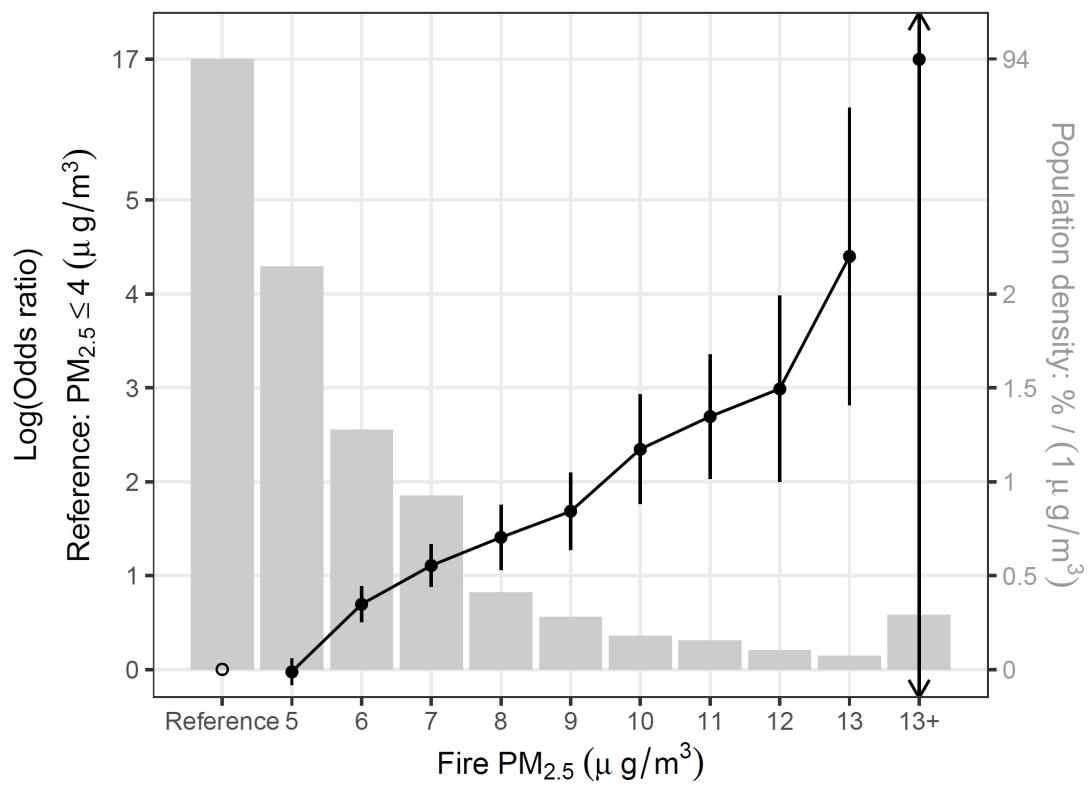
The mean bias is calculated as the difference between the satellite-based PM_{2.5} concentrations and the GEOS-chem simulations. In other words, in the comparison, the satellite-based PM_{2.5} concentrations are treated as the reference values.

† The correlation coefficients are calculated using Pearson's R².

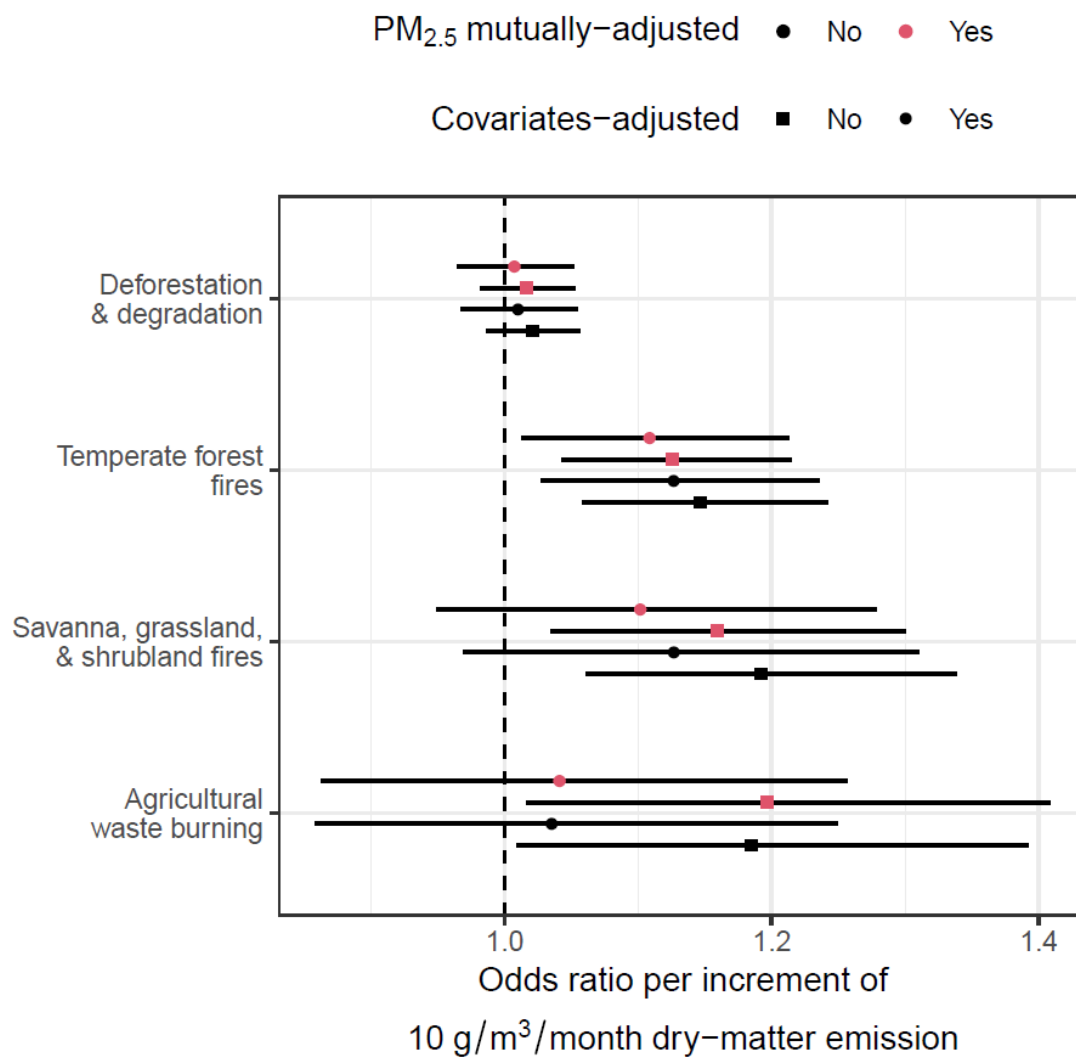
Supplemental Figures



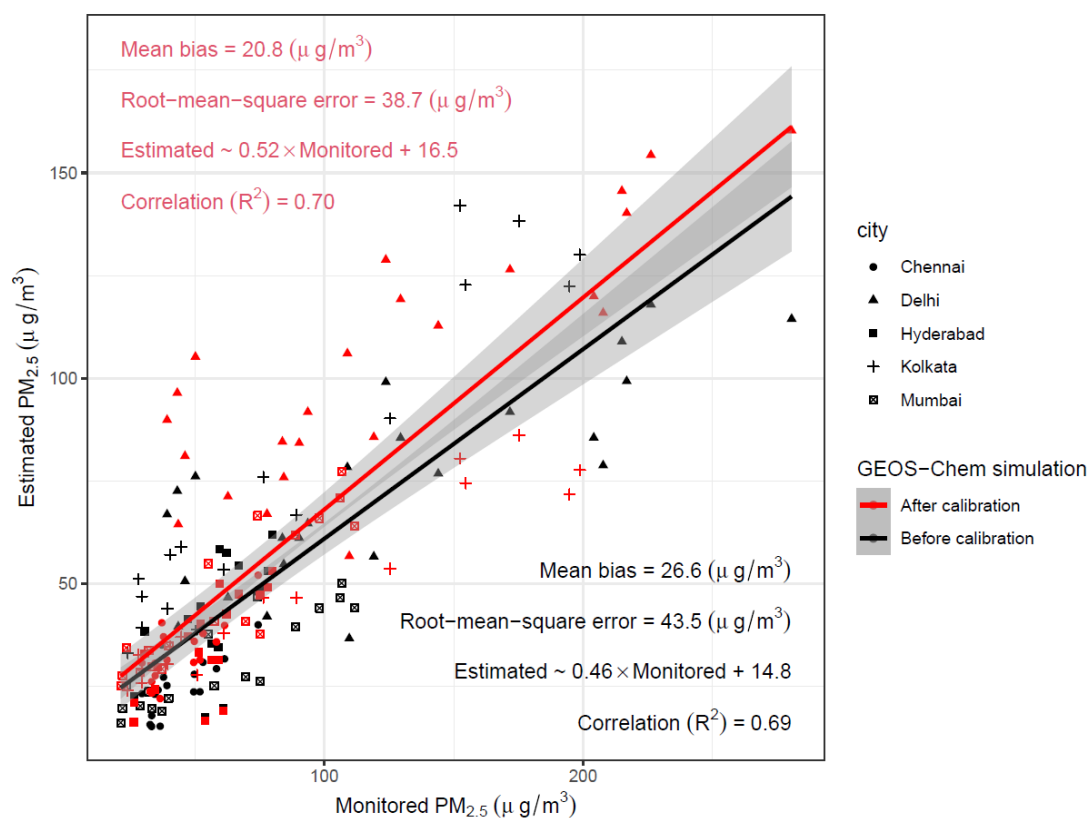
Supplementary Figure 1 Spatial distributions of exposure indicators, averaged during 2000-2014. (a) PM_{2.5} attributable to open fire; (b) total PM_{2.5}; (c) fire emissions and (4) satellite remote sensing of burned area. The maps are generated by the corresponding author (T.X.) using the data from Natural Earth (<https://www.naturalearthdata.com/>). PM_{2.5}: particulate matter with a diameter of less than 2.5 μm .



Supplementary Figure 2 The estimated association between the categorical variable of open fire PM_{2.5} and pregnancy loss. The dots denote point-estimates of the association, the error bars denote the corresponding 95% confidence intervals, and the histograms (bars) denote the distribution of population density. All estimates are derived from the full-adjusted model with 24,876 independent mothers. PM_{2.5}: particulate matter with a diameter of less than 2.5 µm.



Supplementary Figure 3 The estimated associations between fire emissions and pregnancy loss by specific types of fire. The dots denote point-estimates of the associations, and the error bars denote the corresponding 95% confidence intervals. Each type of points (specified by colors and shapes) denotes a separate model with 24,876 independent mothers. PM_{2.5}: particulate matter with a diameter of less than 2.5 μm .



Supplementary Figure 4 Evaluation results for the GEOS-Chem simulated $PM_{2.5}$ concentrations in five Indian cities and monthly scale. The mean bias was calculated by the following equation: mean bias = monitored $PM_{2.5}$ – estimated $PM_{2.5}$. The correlation coefficient is calculated as the Pearson's R^2 . The solid lines denote regressions of the estimates against the monitored values, and the grey ribbons denote their corresponding 95% confidence intervals. $PM_{2.5}$: particulate matter with a diameter of less than 2.5 μm .

Supplementary codes: original R codes for epidemiological analyses

1. R codes for models by different covariate adjustments and exposure indicators

```
load("Lng Fire and PregLoss.RData") # the data is obtained and manually assembled from multiple
surveys from DHS as described in the main-text. Those surveys should be downloaded directly from
DHS website. We cannot re-distribute them.

library(splines)
library(survival)

xs=c("FirePM","FireSat","FireEms","FireEms_AGRI","FireEms_DEFO","FireEms_SAVA","FireEms_TEMF")
lng$PM25gc=lng$PM25gc-lng$FirePM
fs<-c("end~x+strata(v2id)",
      "end~x+PM25gc+strata(v2id)",
      "end~x+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year)+strata(v2id)",
      "end~x+PM25gc+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year)+strata(v2id)")
names(fs)=c("unadjusted","PM25-unadjusted","adjusted","PM25-adjusted")

for(x in xs)
{
  lng$x=lng[,x]
  for(f in fs)
  {
    m<-clogit(as.formula(f),data=lng,iter.max=5000)
    tmpr=rbind(cbind(as.data.frame(summary(m)$coef)[1,]))
    tmpr$model=names(which(fs==f))
    tmpr$x=x
    if(f==fs[1]&x==xs[1]) coef=tmpr else coef=rbind(coef,tmpr)
  }
}
coef$OR=exp(coef$coef*coef$unit)
coef$lo=exp(coef$coef-coef$se(coef)*1.96)
coef$up= exp(coef$coef+coef$se(coef)*1.96)

coef # The model outputs
```

2. R codes for models by modifiers

load("Lng Fire and PregLoss.RData") # the data is obtained and manually assembled from multiple surveys from DHS as described in the main-text. Those surveys should be downloaded directly from DHS website. We cannot re-distribute them.

```
library(splines)
```

```
library(survival)
```

```
xs=c("FirePM","FireSat","FireEms","FireEms_AGRI","FireEms_DEFO","FireEms_SAVA","FireEms_TEMF")
```

```
lng$PM25gc=lng$PM25gc-lng$FirePM
```

```
f<-"end~ef:x+PM25gc+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year)+strata(v2id)"
```

```
fb<-"end~x+PM25gc+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year)+strata(v2id)"
```

```
efs=c("edu","urban","insurance","job","age","anemia","obesity","smoke","marry","gapc")
```

```
for(x in xs[1:3])
```

```
{
```

```
  lng$x=lng[,x]
```

```
  for(ef in efs)
```

```
  {
```

```
    tmp<-subset(lng,v2id%in%subset(lng,lis.na(ef)&end==1)$v2id)
```

```
    m<-clogit(as.formula(f),data=tmp,iter.max=5000)
```

```
    b<-clogit(as.formula(fb),data=tmp,iter.max=5000)
```

```
    tab<-as.data.frame(summary(m)$coef)
```

```
    tab<-tab[grepl("x",rownames(tab)),]
```

```
    ids<-gsub("-", "", fixed=T, substr(rownames(tab), start=3, stop=50))
```

```
tmpr=cbind(x=x,modifier=ef,grp=ids,grp.n=as.numeric(table(tmp$ef)[ids]),grp.p=as.numeric(table(subset(tmp,end==1)$ef)[ids]),tab,LRpvalue=as.numeric(anova(m,b)$"P(>|Chi|)"[2]))
```

```
  if(ef==efs[1]&x==xs[1]) coef=tmpr else coef=rbind(coef,tmpr)
```

```
  }
```

```
}
```

```
coef$OR=exp(coef$coef*coef$unit)
```

```
coef$lo=exp(coef$coef-coef$se(coef)*1.96)
```

```
coef$up= exp(coef$coef+coef$se(coef)*1.96)
```

```
coef # The model outputs
```

3. R codes for nonlinear associations

load("Lng Fire and PregLoss.RData") # the data is obtained and manually assembled from multiple surveys from DHS as described in the main-text. Those surveys should be downloaded directly from DHS website. We cannot re-distribute them.

```
library(splines)
```

```
library(survival)
```

```
lng$PM25gc=lng$PM25gc-lng$FirePM
```

```
f<- "end~pspline(x)+PM25gc+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year) +  
strata(v2id)"
```

```
for(x in c("FirePM"))
```

```
{
```

```
  lng$x=lng[,x]
```

```
  ERF=data.frame(x=seq(0,quantile(lng$x,0.999),length=100),PM25gc=0,TMP=0,  
age=unique(lng$age)[1],HTY=0,bgn.m=1,bgn.year=2000,v2id=unique(lng$v2id)[1])
```

```
  m<-clogit(as.formula(f),data=lng,iter.max=5000)
```

```
  tmp=predict(m,newdata = ERF,type="terms",se=T)
```

```
  ERF$y=tmp$fit[,1]
```

```
  ERF$se=tmp$se.fit[,1]
```

```
  ERF$type=x
```

```
  if(x=="FirePM") ERFs=ERF else ERFs=rbind(ERFs,ERF)
```

```
}
```

```
head(ERF) # The model outputs for nonlinear association
```

```
x="FirePM"
```

```
lng$x=lng[,x]
```

```
lng$x2=cut(lng$x,c(-Inf,4:13,Inf))
```

```
f<- "end~x2+PM25gc+ns(TMP,3)+age+ns(HTY,3)+ns(bgn.m,4)+as.character(bgn.year)+strata(v2id)"
```

```
m<-clogit(as.formula(f),data=lng,iter.max=5000)
```

```
summary(m) # The model outputs for association by categorical variable
```